

Meta-study in the shadow of the future

The effect of continuation probability on strategies

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Motivation beyond repeated games – latent variables are everywhere

Experimental economics

- The fraction of participants who behave according to Expected Utility
- A participant's depth of strategic reasoning in a game (i.e. level- k)
- The fraction of participants using a given strategy in a game (this application)
- A risk premium for a lottery

These are *estimated* rather than *observed*

Motivation beyond repeated games – latent variables are everywhere

Labor economics

- Matching efficiency
- Match quality
- Bargaining power
- Productivity

Macroeconomics

- r^* (the interest rate when the economy is operating a full potential),
- Inflation expectations
- Non-Accelerating Inflation Rate of Unemployment (NAIRU)

Meta-analysis

Aggregating information from more than one study

- This application: aggregating causal effect estimates

Usually done with *observed* outcome variables (e.g. Meager, 2019)

This study: What can I do when I have a latent outcome variable?

- Strategy frequencies

Big picture

What am I studying?

- Direct-response Indefinitely-repeated Prisoner's Dilemma experiments

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What are this project's challenges?

- I don't observe strategies in my meta-dataset
- Not all experiments tell us something about the *causal* effect

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How am I doing this?

- The Strategy Frequency Estimation Method (Dal Bó and Fréchette, 2011; Fudenberg et al., 2012; Dvorak, 2023)
- Meta-analysis (Rubin, 1981), Bayesian hierarchical modelling

Contributions

Methodology

- Meta-analysis of a latent variable: strategies
- Typical meta-analysis: outcome variable is observed

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Indefinitely repeated Prisoner's Dilemma experiments

- Many experiments, but
- As of yet, no **causal** synthesis of the information we got from them
- Exploratory work in Dal Bó and Fréchette (2018)

What do we know so far?

Literature review – Dal Bó and Fréchette (2018)

- Discussion of strategy frequency estimates *qualitatively* across 17 treatments in 6 experiments
 - AD, Grim, and TFT account for most of the data
- Pooled regression models for cooperation rates, 11 experiments
 - Not a causal interpretation (Bland, 2026)
 - **Exploratory** interpretation only

“High cooperation rates will emerge only when the parameters of the repeated game are such that cooperation is very robust to strategic uncertainty.”

Need for **quantitative** and **causal** synthesis of results

Exploratory?

What do I mean by **exploratory**?

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Pooled regression results in a meta-analysis do not have a causal interpretation
(Bland, 2026)

Bland (2026), how *not* to do meta-analysis (of causal effects)

We have decision-level data from many experiments

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We have decision-level data from many experiments

Let's run a regression!

$$\text{Outcome_Variable}_{e,i,t} = \beta_0 + \beta_1 \text{Treatment_Conditions}_{e,i,t} + \epsilon_{e,i,t}$$

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$$\text{Outcome_Variable}_{e,i,t} = \beta_0 + \beta_1 \text{Treatment_Conditions}_{e,i,t} + \epsilon_{e,i,t}$$

But $\hat{\beta}_1$ will be biased if experimenters choose treatment conditions

- A well-known result in panel data econometrics

Experiment fixed effects solve this problem (but might not be efficient)

Experimenters choosing their treatment conditions?

$$\text{Outcome_Variable}_{e,i,t} = \beta_0 + \beta_1 \text{Treatment_Conditions}_{e,i,t} + \epsilon_{e,i,t}$$

All. The. Time!

Experimenters choosing their treatment conditions?

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All. The. Time!

What I worry about specifically: Experimenters choosing treatment conditions in response to beliefs about the error term, $\epsilon_{e,i,t}$

For example:

- I expect a particularly un-cooperative subject pool, so I raise the marginal per-capita return in my public goods game experiment.
- I need to target a particular amount of playing Stag in a Stag Hunt game, so I choose the payoffs to do this knowing my subject pool (G. Fréchette (paraphrased), N American Economic Science Association meeting, 2024)

Experimenters choosing their treatment conditions

To be clear: This is a problem when using *pooled regressions* in *meta-analysis*. It is not a problem with individual experiments.

The take-away: Just because pooled linear regression (or something else) is valid for analyzing the individual experiments does not mean that it is OK to analyze the meta-dataset like this (and still have a causal interpretation).

Preview of results

Increasing the continuation probability δ in an indefinitely-repeated Prisoner's Dilemma:

- Significantly¹ increases first-round cooperation (observable outcome variable)

¹Based on a 95% Bayesian credible region, 2.5th-97.5th percentile, not including zero

Preview of results

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You will likely find these effects in individual experiments, but they will be harder to detect.

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The indefinitely repeated Prisoner's Dilemma

	C	D
C	1 1	$-l$ $1 + g$
D	$1 + g$ $-l$	0 0

Table: The normalized repeated Prisoner's Dilemma (Dal Bó and Fréchette, 2018)

Play again (and again, etc.) with *continuation probability* $\delta \in (0, 1)$

g : Gain from defecting when opponent plays C

l : Loss from cooperating when opponent plays D

Strategies in indefinitely repeated games

Strategy	Abbreviation	Description
Always defect	AD	Always play D
Always cooperate	AC	Always play C
Grim trigger	Grim	Begin with playing C. Only play C in subsequent rounds if all past actions have been C, otherwise play D
Lenient Grim 2	Grim2	Begin by playing C. Switch to playing D forever if at least one D action is played in two consecutive rounds.
Tit for Tat	TFT	Begin with playing C. Copy opponent's previous action in subsequent rounds
Suspicious Tit for Tat	STFT	Begin by playing D. Copy opponent's previous action in subsequent rounds.
2 Tits for Tat	2TFT	Begin by playing C. Punish opponent's defection in previous round with two consecutive round of defection. Otherwise play C.
Tit for 2 Tats	TF2T	Begin by playing C. Punish opponent for playing D in two consecutive rounds by playing D. Otherwise play C.

The problem with strategies

In indefinitely repeated games, the strategy space is indefinitely large

In **direct response** experiments, **we do not observe strategies**

- Alternative: strategies are elicited (e.g. Romero and Rosokha, 2018; Dal Bó and Fréchette, 2019; Gill and Rosokha, 2024)

Instead, we observe **actions**

Can think of actions as the result of two strategies playing against each other

But strategies are important!

A **fundamental** part of game theory

Crucial to understanding how/why/if cooperation can/will occur

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Meta-analysis

What can we learn from the existing literature?

Search existing literature for IRPD experiments with:

- Perfect monitoring
- Deterministic, symmetric payoffs
- Strictly positive continuation probability δ
- Direct response method used

Include all treatments that satisfy these conditions

In the existing literature, I found 27 experiments with available decision-level data

How did I search for data?

- My own literature review (ongoing)
- Post on the Economic Science Association “discussion” Google group (February 2025)
- Twitter and BlueSky posts (February 2025)

All of these were fruitful! Data *are* available upon request!

Some working papers included, only with permission of all authors

- E.g. Bland et al. (2023), Kasberger et al. (2023)

Where did the data come from?

A total of 27 experiments so far

Mostly from Fudenberg and Rehbindler (2024), which built on Dal Bó and Fréchette (2018).

Removed one duplicate experiment from this dataset.

To this I added Bland et al. (2023), Proto et al. (2019), Aoyagi et al. (2024), Cason and Mui (2019), Casoria and Ciccone (2021), Kasberger et al. (2023), Kölle et al. (2023), Lambrecht et al. (2024), Maggian and Spinola (2024), Proto et al. (2021), Romero and Rosokha (2018), and Stringhi et al. (2025)

However: Not all of these can answer my research questions

What questions can the data answer?

Here, I want to make *causal* statements about game parameters g , l , and δ .

Bland (2026, again): Only include an experiment in a meta-analysis if it can *individually* answer the research question

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No experiment allows me to estimate the causal effect of l (loss from cooperating when opponent defects)

Only two experiments allow me to estimate the causal effect of g (gain from defecting)

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No experiment allows me to estimate the causal effect of l (loss from cooperating when opponent defects)

Only two experiments allow me to estimate the causal effect of g (gain from defecting)

Eleven experiments allow me to estimate the causal effect of continuation probability δ

- Focus on these

The useful data

Experiment	g	l	δ
Arechar, Kouchaki, and Rand 2018	0.333	0.333	0.125 0.875
Bland, Bochet, Nikiforakis, and Zhang 2023	0.800	0.200	0.100 0.800
	1.333	1.000	0.100 0.800
Blonski, Ockenfels, and Spagnolo 2011	0.500	3.500	0.750 0.875
	2.000	2.000	0.500 0.750 0.875
Casoria and Ciccone 2021	2.571	3.000	0.350 0.750
Dal Bo and Frechette 2011a	0.087	0.565	0.500 0.750
	0.667	0.867	0.500 0.750
	2.571	1.857	0.500 0.750
Dal Bo and Frechette. 2019	0.087	0.565	0.500 0.750
	2.571	1.857	0.500 0.750 0.900
Ghidoni and Suetens 2021	0.087	0.565	0.500 0.750
	0.667	0.867	0.500 0.750
	2.571	1.857	0.500 0.750
Kasberger, Martin, Normann, and Werner 2023	0.087	0.565	0.900 0.950
	0.667	0.867	0.900 0.950
Kölle, Quercia, and Tripodi 2025	2.000	2.000	0.600 0.800
Proto, Rustichini, and Sofianos 2019	0.087	0.565	0.500 0.750
Romero and Rosokha 2023	2.571	1.857	0.750 0.950

Research questions

Research Question

What is the effect of continuation probability δ on first-round cooperation?

- *Relates to an **observable** outcome variable*
- *Usually how direct-response IRPD experiments are analyzed*

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Research Question

What is the effect of continuation probability δ on the fraction of participants using each strategy?

- *Relates to **latent** outcome variables*

Research questions

Research Question

What is the effect of continuation probability δ on the fraction $\frac{\text{TFT}}{\text{TFT}+\text{Grim}}$?

Related to findings of Dal Bó and Fréchette (2019) and Romero and Rosokha (2023) in **strategy method** experiments that this ratio is increasing in δ

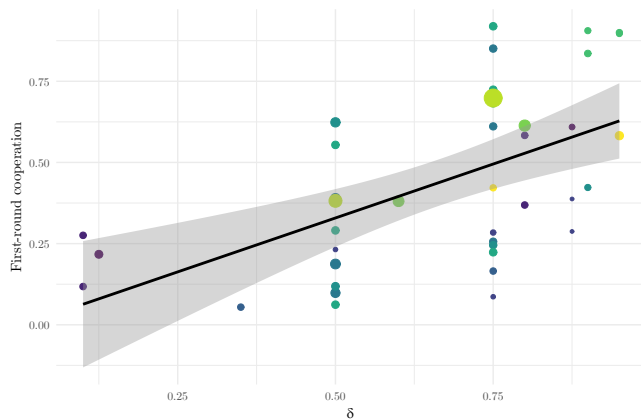
Research questions

Research Question

What is the effect of continuation probability δ on cooperative, forgiving, and lenient strategies?

- **Cooperative:** Plays C at the beginning of the supergame
- **Forgiving:** Can return to playing C after a punishment phase
- **Lenient:** Requires at least two defections before punishment is triggered.

The useful data – first-round cooperation



experiment

- Arechar, Kouchaki, and Rand 2018
- Bland, Bochet, Nikiforakis, and Zhang 2023
- Blonski, Ockenfels, and Spagnolo 2011
- Casoria and Ciccone 2021
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- Proto, Rustichini, and Sofianos 2019
- Romero and Rosokha 2023

decisions

- 5000
- 10000
- 15000

But we can't just draw a line through this!

The basic OLS line would make comparisons (Bland, 2026):

- 1 Between treatments with the same stage-game payoffs
- 2 Between treatments with different stage-game payoffs
- 3 Between different experiments

(1) is good, (2) is bad, (3) is very bad

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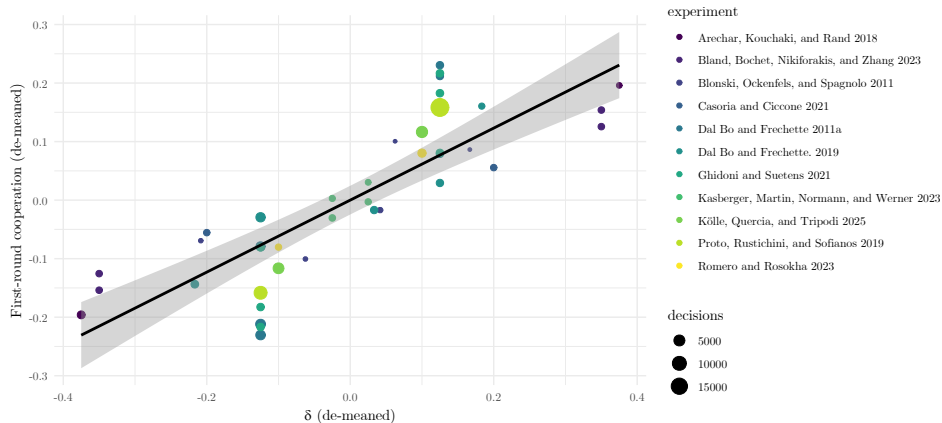
- 1 Between treatments with the same stage-game payoffs
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(1) is good, (2) is bad, (3) is very bad

Solution: Only make comparisons between treatments in the same experiment with the same stage-game payoffs

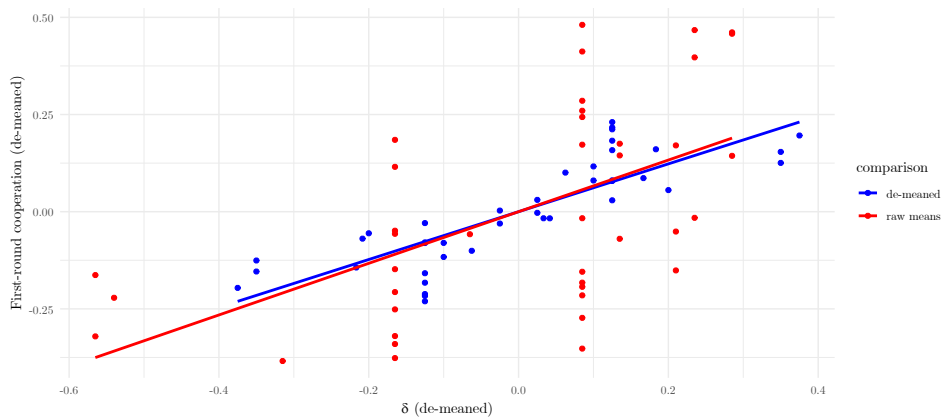
- We can do this by de-meaning δ and cooperation rates within these groups of treatments
- Equivalent to experiment $\times$ stage-game payoffs fixed effects

The useful data – first-round cooperation, valid treatment comparisons



The useful data – first-round cooperation, comparing the plots

Raw means slightly over-estimate the effect of δ



But this is still too “exploratory” for me

- OLS estimator weights experiments arbitrarily
 - Some treatment comparisons and experiments may be more informative than others
- Treatment effects are likely heterogeneous between experiments

Hold tight for the methodology!

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Overview of methodology

- 1 Estimate strategy frequencies (RQ2-4 only)
- 2 Estimate causal effect of δ using only valid treatment comparisons
- 3 Aggregate up to an experiment-averaged causal effect estimate
- 4 Aggregate information across experiments

1. Strategy frequency estimation method (SFEM)

Problem 1: There are infinitely many strategies that players could be using

- **Solution:** Assume a finite list of possible strategies being used

Problem 2: I don't observe strategies in these experiments

- **Solution:** Estimate the fraction of participants using each strategy
- Finite mixture model

1. SFEM – A finite list of strategies

Strategy	Abbreviation	Description
Always defect	AD	Always play D
Always cooperate	AC	Always play C
Grim trigger	Grim	Begin with playing C. Only play C in subsequent rounds if all past actions have been C, otherwise play D
Lenient Grim 2	Grim2	Begin by playing C. Switch to playing D forever if at least one D action is played in two consecutive rounds.
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1. SFEM – estimation

Data:

- $y_{i,s}$ is the number of times player i followed strategy s
- n_i is the number of decisions

Parameters:

- ρ_s is the fraction of participants using strategy s . $\rho \in \Delta^S$
- $\epsilon \in (0, 0.5)$ is the *tremble probability*

Likelihood function:

$$\log p(y \mid \rho, \epsilon) = \sum_{i=1}^N \log \left(\sum_{s=1}^S (1 - \epsilon)^{y_{i,s}} \epsilon^{n_i - y_{i,s}} \rho_s \right)$$

Maximum likelihood estimation implemented in *Stan* (Carpenter et al., 2017)

2. Causal effect within valid treatment comparison groups

Only valid to compare two or more treatments if:

- 1 All treatments come from the same experiment
- 2 All treatments only vary by δ (i.e. all have the same g and l)

2. Causal effect within valid treatment comparisons

Suppose set of treatments J is a valid treatment comparison group

We can estimate the causal effect of δ on first-round cooperation $c_{e,j,i,t}$ using linear regression:

$$\hat{\tau}_{e,J,c} = \frac{\sum_{j \in J} \sum_{i,t \in j} c_{e,j,i,t} (\delta_{e,j} - \bar{\delta}_{e,J})}{\sum_{j \in J} \sum_{i,t \in j} (\delta_{e,j} - \bar{\delta}_{e,J})^2}$$

Similarly for strategies:

$$\hat{\tau}_{e,J,s} = \frac{\sum_{j \in J} \hat{\rho}_{e,j,s} (\delta_{e,j} - \bar{\delta}_{e,J})}{\sum_{j \in J} (\delta_{e,j} - \bar{\delta}_{e,J})^2}$$

3. Experiment-averaged causal effect estimates

We might have more than one valid treatment comparison group

For example: two in Bland et al. (2023), three in Dal Bó and Fréchette (2011)

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Any weighted average of the within-group estimates preserves unbiasedness

$$\hat{\tau}_{e,s} = \sum_{J \in \mathcal{J}_e} w_{e,J,s} \hat{\tau}_{e,J,s}$$

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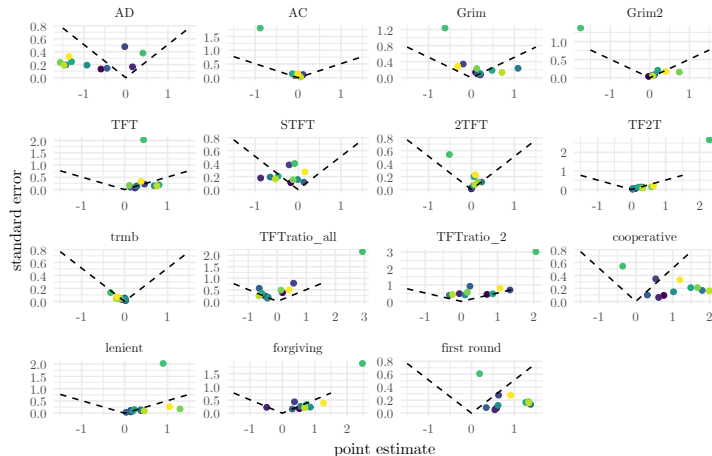
Any weighted average of the within-group estimates preserves unbiasedness

$$\hat{\tau}_{e,s} = \sum_{J \in \mathcal{J}_e} w_{e,J,s} \hat{\tau}_{e,J,s}$$

But we can minimize the variance with the following:

$$w_{e,J,s} = \frac{V(\hat{\tau}_{e,J,s})^{-1}}{\sum_{K \in \mathcal{J}_e} V(\hat{\tau}_{e,K,s})^{-1}}$$

3. Experiment-averaged causal effect estimates



experiment

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4. Meta-analysis

Up to this point: 11 experiment-specific estimates of the causal effects of δ on first-round cooperation and strategies, and their associated standard error estimates

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Meta-analysis: What do we learn from all of these experiments together?

- What is “the” causal effect at the population level?
- Account for heterogeneity in effects at the experiment level
- How generalizable are the experiments’ results?

4. Meta-analysis – The Rubin (1981) model

Two sources of variation in the experiments' estimates:

- 1 Sampling variation
- 2 Heterogeneous true effects

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$$\hat{\tau}_{e,s} \sim N(\mu_{e,s}, \widehat{\text{se}}(\hat{\tau}_{e,s})^2)$$

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Between-experiment variation Experiment-specific effects are distributed around a population mean $\bar{\mu}_s$:

$$\mu_{e,s} \sim N(\bar{\mu}_s, \bar{\sigma}_s^2)$$

4. Meta-analysis – implementation

$$\hat{\tau}_{e,s} \sim N(\mu_{e,s}, \widehat{\text{se}}(\hat{\tau}_{e,s})^2), \quad \mu_{e,s} \sim N(\bar{\mu}_s, \bar{\sigma}_s^2)$$

Bayesian estimation with hyper-prior:

$$\bar{\mu}_s \sim N(0, 0.25^2), \quad \bar{\sigma}_s \sim \text{Cauchy}^+(0, 0.1)$$

MCMC sampling in *Stan* (Carpenter et al., 2017)

4. Meta-analysis – questioning the assumptions

How appropriate is it to assume things are normally distributed?

How appropriate are the hyper-priors?

How appropriate is the choice of data?

4. Meta-analysis – the normal assumptions

Experiment level:

$$\hat{\tau}_{e,s} \sim N(\mu_{e,s}, \widehat{\text{se}}(\hat{\tau}_{e,s})^2)$$

- These estimators are all asymptotically normal

4. Meta-analysis – the normal assumptions

Experiment level:

$$\hat{\tau}_{e,s} \sim N(\mu_{e,s}, \widehat{\text{se}}(\hat{\tau}_{e,s})^2)$$

- These estimators are all asymptotically normal

Population level:

$$\mu_{e,s} \sim N(\bar{\mu}_s, \bar{\sigma}_s^2)$$

- Much more heroic
- Probably need more experiments to relax this assumption (then could use Student's t -distribution)

4. Meta-analysis – choice of hyper-prior

$$\bar{\mu}_s \sim N(0, 0.25^2), \quad \bar{\sigma}_s \sim \text{Cauchy}^+(0, 0.1)$$

- Prior 95% credible region for $\bar{\mu}_s$ is $(-0.49, 0.49)$
 - Centred on zero
 - For an effect size where the cause is $\delta \in (0, 1)$
 - Outcome must be in the range $(-1, 1)$
- For $\bar{\sigma}_s$, $p(\bar{\sigma}_s < 1.27) \approx 0.95$
- Prior sensitivity analysis

4. Meta-analysis – choice of data

Re-run analysis leaving out one experiment at a time

Used to assess influence of each experiment

Results are robust to this

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Results – first-round cooperation

Result

A 10 percentage point increase in continuation probability δ on average causes a 6.4 percentage point increase in first-round cooperation. We can expect an effect in this direction in 91 percent of experiments.

Strategy	$\bar{\mu}$ (mean of effect)	$\bar{\sigma}$ (std of effect)	probability of positive effect
first round	0.640 (0.153)*	0.444 (0.148)	0.910 (0.094)

Results – strategies

Result

Increasing δ on average causes a decrease in the use of AD and STFT strategies, and increases in the Grim, TFT, and TF2T strategies

Strategy	$\bar{\mu}$ (mean of effect)	$\bar{\sigma}$ (std of effect)	probability of positive effect
2TFT	0.025 (0.032)	0.040 (0.034)	0.688 (0.247)
AC	0.033 (0.028)	0.035 (0.028)	0.808 (0.217)
AD	-0.436 (0.195)*	0.737 (0.218)	0.266 (0.116)
Grim	0.228 (0.101)*	0.275 (0.116)	0.804 (0.135)
Grim2	0.133 (0.073)	0.172 (0.081)	0.789 (0.131)
STFT	-0.222 (0.097)*	0.278 (0.097)	0.209 (0.123)
TF2T	0.192 (0.063)*	0.167 (0.059)	0.871 (0.092)
TFT	0.350 (0.077)*	0.189 (0.082)	0.947 (0.072)
ϵ	-0.074 (0.031)*	0.085 (0.027)	0.193 (0.103)

Results – strategy ratios

Result

The effect of continuation probability on the ratio $\frac{\text{TFT}}{\text{TFT}+\text{Grim}}$ is mixed.

Strategy	$\bar{\mu}$ (mean of effect)	$\bar{\sigma}$ (std of effect)	probability of positive effect
$\frac{\text{TFT}}{\text{TFT}+\text{Grim}}$	0.175 (0.138)	0.126 (0.127)	0.834 (0.239)
$\frac{\text{all TFT}}{\text{all TFT}+\text{Grim}}$	-0.276 (0.093)*	0.085 (0.082)	0.037 (0.091)
cooperative	0.518 (0.228)*	0.832 (0.260)	0.746 (0.123)
forgiving	0.406 (0.125)*	0.331 (0.148)	0.878 (0.113)
lenient	0.318 (0.109)*	0.350 (0.110)	0.820 (0.107)

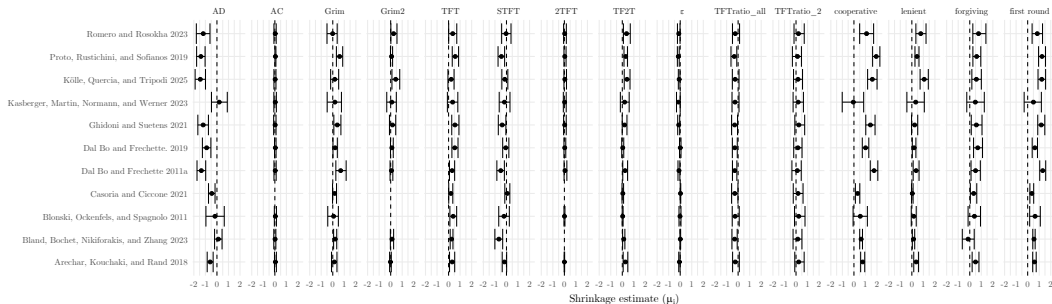
Results — cooperative, forgiving and lenient

Result

As δ increases, players typically use more cooperative, forgiving, and lenient strategies

Strategy	$\bar{\mu}$ (mean of effect)	$\bar{\sigma}$ (std of effect)	probability of positive effect
$\frac{\text{TFT}}{\text{TFT}+\text{Grim}}$	0.175 (0.138)	0.126 (0.127)	0.834 (0.239)
$\frac{\text{all TFT}}{\text{all TFT}+\text{Grim}}$	-0.276 (0.093)*	0.085 (0.082)	0.037 (0.091)
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Experiment-specific (shrinkage) estimates: $\mu_{e,s}$



Error bars show 95% Bayesian credible regions, 2.5th-97.5th percentiles

The biggest exception for these estimates

Kasberger et al. (2023)

No significant experiment-specific effect for 1st-round cooperation, AD, Grim, and TFT

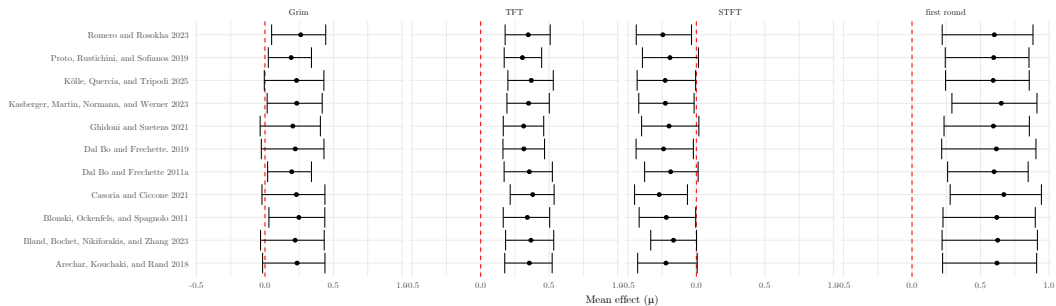
- Four treatments
- $\delta = 0.90$ and $\delta = 0.95$
- My model assumes the causal effect is constant with respect to δ
- In reality, it is unlikely that participants adjust their strategies much in this range.

Robustness checks

How robust are the results to:

- 1 Studies included?
- 2 Hyper-prior scale?

Sensitivity to studies included



Sensitivity to hyper-prior

Inflating prior scale by a factor of ten (showing posterior means of μ only)

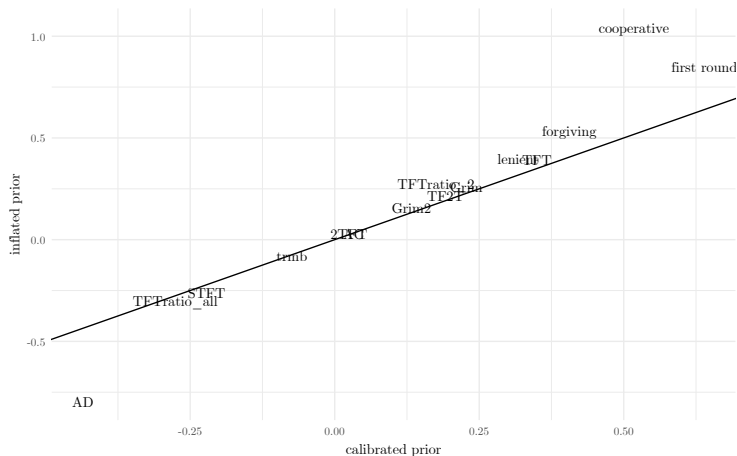


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Findings

An increase in continuation probability δ causes a significant

- Increase in first-round cooperation
- Increase in the use of the Grim, TFT, and TF2T strategies
- Decrease in the use of the AD and STFT strategies

Results mostly replicate within experiments, but more noisy

Contributions

For indefinitely-repeated Prisoner's Dilemma experiments

- Aggregation of **causal** effects using multiple experiments
- Estimates of the effect of game parameters on strategies (in direct-response experiments)

For meta-analysis methodology

- Meta-analysis of a latent variable
- If you can estimate the latent variable, the steps are fairly straightforward
- Latent variable must have an intrinsic scale (Stoetzer et al., 2024)

Remaining gaps in the literature

Very few experiments can speak to the effect of (normalized) stage-game payoffs (g, l)

Derived quantities from these that theory says should be important

- Critical δ for supporting cooperation (a function of g)
- Size of the basin of attraction for Grim risk-dominating AD (a function of g , l , and δ)

Thank you!

Questions? Comments?

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